From models to agents and robots

Technological China · Class 2 · Intelligent systems and production.

Sergio Gevatschnaider
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The class connects technical capability to outcomes

A 60-minute core route with extensions for later study.

01	**Foundations · 8 min**	Agent, environment and tool use.
02	**Models and platforms · 12 min**	DeepSeek, Qwen and distribution.
03	**Robotics and videos · 17 min**	Physical architecture and observable evidence.
04	**Lab · 15 min**	Energy, operational success and cost.
05	**Discussion · 8 min**	Competitiveness, work and opportunities.

Economic value appears in correct, repeatable tasks

A model capability needs integration and operating conditions.

Capability

Solve a task under defined conditions.

→

Reliability

Repeat it with measured errors and limits.

→

Economics

Create net value after integration and operation.

Model, chatbot, agent and robot describe different functions

They can be combined within one product.

Model

Component that predicts or generates outputs.

Chatbot

Conversational interface with a user.

Agent

System that observes, decides and acts.

Robot

Machine that senses and acts physically.

Five layers connect research to production

A company’s advantage also depends on suppliers and channels.

01	**Models**	DeepSeek, Qwen and other families.
02	**Agent systems**	Orchestration, tools and evaluation.
03	**Cloud and distribution**	Access to compute, users and services.
04	**Infrastructure**	Chips, memory, interconnects and energy.
05	**Physical world**	Factories, sensors, robots and support.

Strategies combine models, platforms and infrastructure

These examples illustrate different ecosystem positions.

DeepSeek

Research on models and compute efficiency.

Alibaba / Qwen

Models and tools connected to cloud and distribution.

Huawei

Sector models and computing infrastructure.

DeepSeek-V3 activates a fraction of its parameters per token

MoE routes processing through selected experts.

MoE

671 billion

Total parameters of V3 in the report.

37 billion

Parameters activated per token, according to the report.

Implication

Less activation does not remove memory, communication or complexity.

R1 and R1-Zero use different training paths

Both start from a pretrained base.

Pretrained base

Capabilities acquired from large corpora.

R1-Zero

Reinforcement learning without preliminary SFT in that experiment.

R1

Cold-start data, reinforcement stages and additional tuning.

The reported V3 cost has a specific scope

The figure covers official training compute under an assumed rental price.

US$5.576m

What it calculates

2.788 million H800 hours × US$2 per hour.

What it excludes

Prior research and ablation experiments.

What it does not measure

Total company costs or the full development of R1.

Model selection requires a task-level evaluation

A general benchmark does not replace a use-case test.

Quality

Success on representative data and difficult cases.

Latency and cost

Total time, tokens, tools and retries.

Access and license

Weights, API and terms for the chosen variant.

Operations

Privacy, language, availability and support.

Qwen and Qwen-Agent occupy different levels

System capability depends on configured components.

Qwen

Model family with variants and modalities.

Qwen-Agent

Framework for integrating models and tools.

Application

A product’s interface, permissions, data and services.

Distribution can turn capability into adoption

A channel reduces friction when it solves an existing user need.

Access

Existing users and service discovery.

Integration

Connected data, processes and tools.

Retention

Recurring quality and switching costs.

Tools make parts of execution verifiable

The log should distinguish request, result and validation.

Model

Requests a function and proposes arguments.

→

Application

Validates scope, arguments and permissions.

→

Tool

Calculates, searches, opens or modifies a resource.

→

Verification

Checks the result before concluding.

Supplier research requires normalization before comparison

A fictional exercise on sources and comparison criteria.

01	**Same task**	Handle the same part under the same requirements.
02	**Same measure**	Per-hand, total or recommended payload: define scope.
03	**Same evidence**	Source, date, variant and operating test.
04	**Output**	Table with uncertainty and testing needs.

A persuasive demo needs an evidence record

Assess what actually ran and could be checked.

Claim

“I compared the three best suppliers.”

Record

Opened sources, extracted fields and operations.

Validation

Selection criteria and rejected alternatives.

Embodied intelligence connects perception to physical action

具身智能 · jùshēn zhìnéng.

具身智能

jùshēn zhìnéng

Perception

Interpret objects, contact and position.

Decision

Choose an action based on task and state.

Control

Execute motion and correct deviations.

Robot architecture separates goals from control

Layers address different problems and timescales.

01	**Task**	Select the object and destination.
02	**Perception and state**	Estimate pose, obstacles and contact.
03	**Policy and trajectory**	Choose actions or motion references.
04	**Control and actuators**	Regulate forces and motion with feedback.

A VLA conditions actions on vision and instructions

Vision–Language–Action combines modalities for a robotic task.

Input

Image, instruction and available state.

→

VLA policy

Prediction of actions or action sequences.

→

Execution

Conversion to motion through the control system.

Robots learn from demonstrations, interaction and simulation

Each data source brings different information and limitations.

Teleoperation

A human demonstrates a task.

Imitation

A policy learns observation–action relationships.

Simulation and RL

Scenario exploration with a defined objective.

Physical tests

Validation with real contact, failures and variability.

Simulation leaves a gap to the real world

A simulator approximates friction, contact, sensors and dynamics.

What changes

Surfaces, lighting, mass and wear.

How to narrow the gap

Scenario variation, calibration and adaptation.

What to check

Success and safety under unseen conditions.

Industrial policy needs implementation indicators

An announced target and deployed capability require different evidence.

Inputs

Funding, talent and infrastructure.

Activities

Labs, training and industrial integration.

Outcomes

Adoption, quality, productivity and safety.

Unitree, UBTECH and XPENG show three starting points

The comparison identifies emphasis, not a universal winner.

Unitree

Hardware, mobility and research platforms.

UBTECH

Integration of humanoids into industrial processes.

XPENG

Transfer of capabilities from vehicles and manufacturing.

G1 illustrates mobility and hardware configuration

The manufacturer’s specifications describe conditional capabilities.

G1

≈ 132 cm

Declared G1 height.

≈ 35 kg

Declared weight depending on configuration.

Operating question

How many valid tasks does it complete without intervention?

Walker S2 motivates a discussion of uptime

UBTECH declares autonomous battery swapping within three minutes.

≤ 3 min

Energy

Replenishment reduces one source of downtime.

Reliability

Failures, parts, maintenance and assistance still matter.

Metric

Availability = available hours / planned hours.

IRON links automotive manufacturing to robotics

Company announcement dated September 8, 2026.

01	**Reported as completed**	Commissioning a line and producing a robot.
02	**Declared target**	Mass production targeted for late 2026.
03	**Commercial plan**	Launch and deliveries planned for 2027.

A useful comparison retains task, version and conditions

Missing data belongs in the comparison.

Dimension	Evaluation question
Locomotion	Which surface and what stability?
Manipulation	Which object, precision and repeatability?
Autonomy	How much intervention and remote assistance?
Operations	What uptime and support?
Economics	What cost per correct task?

Videos have a dedicated page for observation and discussion

Each demonstration includes a question and an evidence limitation.

G1 · dynamic control

Observe balance and motion recovery.

IRON · locomotion

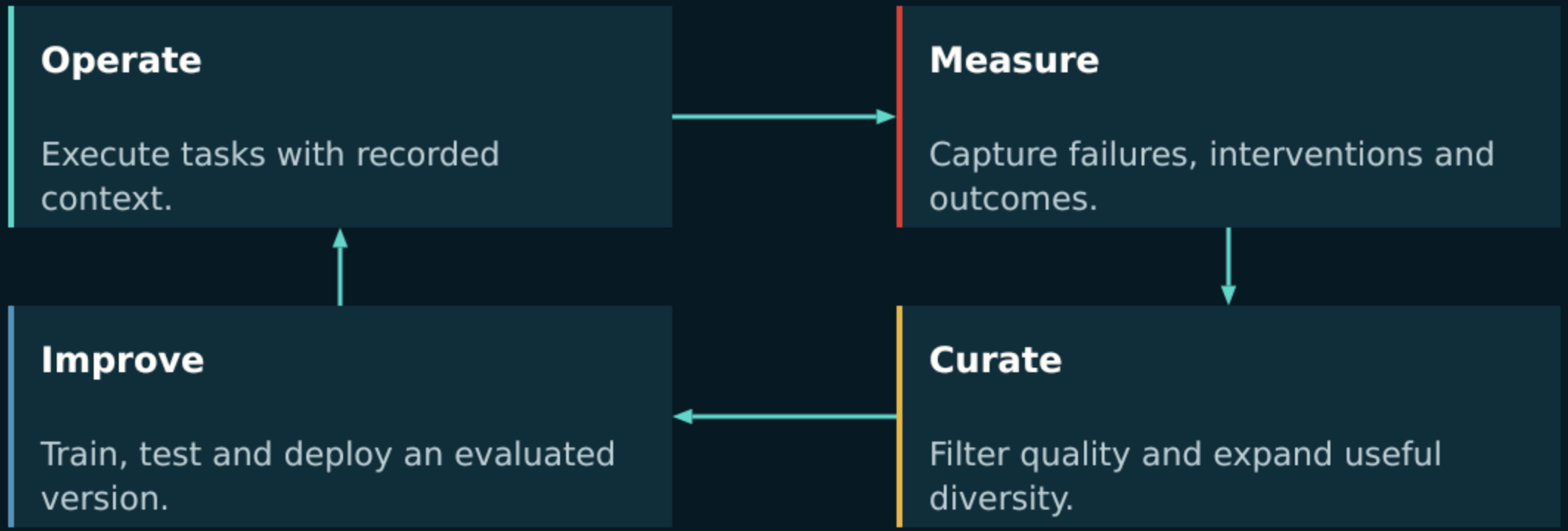
Distinguish human-like appearance from autonomy.

Walker S2 · battery

Ask what is missing to assess uptime.

A data flywheel requires usage to improve the system

Interaction data connect perception, action and outcome.



Data quality shapes transferable learning

More recorded hours do not necessarily mean more useful information.

Coverage

Parts, lighting, shifts and exceptions.

Labeling

Distinguish attempt, success, failure and recovery.

Rights

Permitted access and use of industrial data.

Transfer

Adaptation to different hardware and environments.

Autonomy and economic evidence are separate axes

Quadrants describe evidence types, not company rankings.

More economic evidence

Less

Profitable narrow case Narrow task and measured costs.	**Validated pilot** Autonomy and cost assessed.
Prepared demonstration Little operating evidence.	**Autonomous demo** Observed capability, open economics.

Less autonomy

More autonomy

Gross throughput and valid tasks measure different things

Success rate and availability modify useful output.

$$q_v = q \times A \times r$$

q · gross tasks / h

Rate during an operating hour.

A · availability

Available share of scheduled time.

r · success rate

Share of tasks completed correctly.

RobotLab retains a simple energy model

The original equations are teaching assumptions.

$$E(s) = (150 + 100s^2 + 80w) / (30swe)$$

Throughput

$q = 30 \times s \times w$ tasks per operating hour.

Power

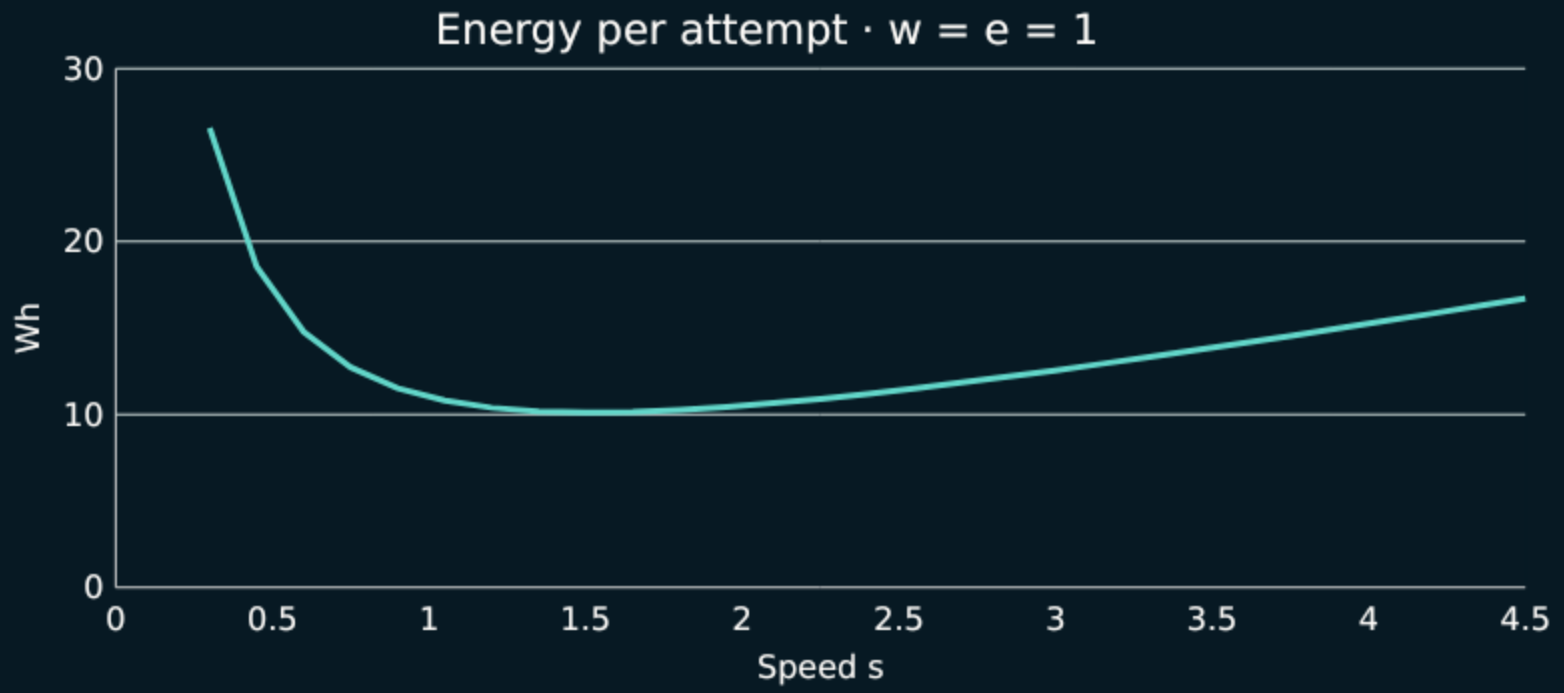
$W = (150 + 100s^2 + 80w) / e$.

Energy intensity

$E = W / q$ Wh per attempt.

Minimum energy occurs at an intermediate speed

At fixed w, energy per attempt first falls and then rises.



s* ≈ 1,52

Energy minimum

Efficiency e rescales the curve.

This is not a factory cost optimum.

Battery energy imposes an exact time constraint

The simulation stops energy use and task accumulation at depletion.

$$t = B / W$$

Runtime

$t = B / W$, with B in Wh and W in watts.

State

$B_{t+\Delta} = \max(0, B_t - W \times \Delta h)$.

Comparison

A larger battery changes runtime, not Wh per attempt.

Energy efficiency alone does not determine useful cost

The extension adds availability, success rate and annual costs.

Annual valid tasks

$Q_v = H \times A \times q \times r$.

Energy per valid task

$E_v = W / (q \times r)$, during operation.

Unit cost

Total annual cost / Q_v.

Total cost includes integration and operation

Compare technologies by valid task and economic horizon.

Initial investment

Equipment, installation, adaptation and engineering.

Annual fixed cost

Annualized capital, support, licenses and maintenance.

Variable cost

Energy, consumables and intervention per attempt.

A fixed task can favor conventional automation

Hypothetical example with annual costs already calculated.

Alternative	Annual cost	Valid tasks	Cost / task
Humanoid A	US$80,000	40,000	US$2.00
Arm B	US$56,000	70,000	US$0.80

Usage volume spreads fixed costs

Lower variable cost offsets investment only with enough activity.

$$Q^* = (F_a - F_\beta) / (v_\beta - v_a)$$

Alternative A

Cost = $F_a + v_aQ$.

Alternative B

Cost = $F_\beta + v_\beta Q$.

Crossover

$Q^* = (F_a - F_\beta) / (v_\beta - v_a)$.

Automation reorganizes tasks and skills

Substitution and complementarity can occur within one occupation.

Changing tasks

Transport, sorting or repetitive inspection.

Complementary tasks

Diagnostics, integration, quality and supervision.

Conditions

Training, work organization and adoption.

“New quality productive forces” links innovation and organization

新质生产力 · xīn zhì shēngchǎn lì.

新质生产力

xīn zhì shēngchǎn lì

Technology

Research and process improvements.

Capabilities

Talent, suppliers and infrastructure.

Outcomes

Observable quality and productivity.

Bottlenecks appear throughout the system

A component’s efficiency depends on complementary assets.

Compute

Chips, memory and interconnects.

Engineering

Actuators, precision, sensors and calibration.

Operations

Energy, maintenance and local support.

Market

Demand, integration and ability to pay.

Argentina can supply services that make technology usable

Opportunities to assess, not a guaranteed demand forecast.

Evaluation

Comparable model and task testing.

Integration

Connections to client systems and processes.

Data

Curation, evaluation and bilingual documentation.

Economics

Total cost, scenarios and process redesign.

Chinese vocabulary helps interpret technology discussions

Research, engineering and policy concepts.

具身智能	jùshēn zhìnéng	Embodied intelligence
数据飞轮	shùjù fēilún	Data flywheel
端到端	duān dào duān	End-to-end
新质生产力	xīn zhì shēngchǎn lì	New quality productive forces
机器人红利	jīqìrén hónglì	Robotic dividend (metaphor)

The workshop asks for a decision supported by assumptions

Compare two scenarios using the lab.

01	**Hypothesis**	“More speed reduces cost per valid task.”
02	**Experiment**	Change speed and then success rate, one variable at a time.
03	**Evidence**	Save A and B, export CSV and explain the result.
04	**Limit**	State what must be measured before applying it in a factory.

The portal connects decks, lab and videos

Local study materials with external links for further reading.

Presentations

Editable PPTX, PDF and a reader with notes and search.

Lab

Energy, operations and an agent workflow.

Videos

Dedicated page with access to the repository VideoLab.

Teaching guide

60-minute route, answers and extensions.

Evidence connects models, agents and productivity

The relevant outcome is a useful, repeatable and economically justified task.

01

Model

What it can produce.

02

Agent

What it can execute and verify.

03

Robot

What it can perform physically.

04

Economics

What value remains after all costs.